

Functional groups

HYDROCARBONS CAN REACT WITH OTHER ELEMENTS.

SEE ALSO

◀ 28–29 Respiration

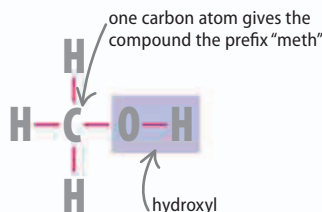
◀ 78–79 Cycles in nature

◀ 144–145 Acids and bases

These “functional groups” of additional elements dominate the compound’s chemical behavior.

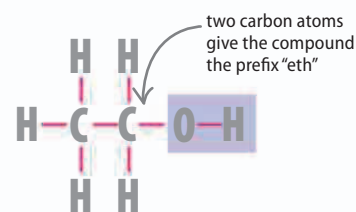
Alcohols

These are organic molecules in which an oxygen and hydrogen (–OH) is added to the carbon chain, in the place of a hydrogen atom. Ethanol—the alcohol with two carbon atoms—is the compound in alcoholic drinks. It is produced by natural fermentation processes and can be metabolized by the body. However, all other alcohols are much more poisonous.



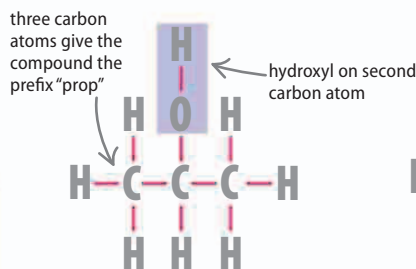
△ Methanol

This simplest alcohol is used as an antifreeze and solvent.



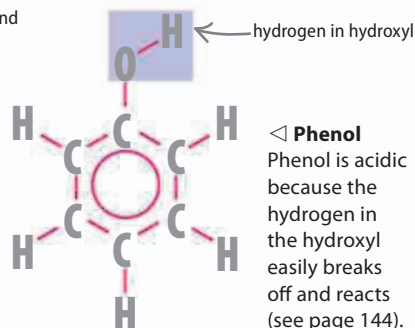
△ Ethanol

This alcohol is found in beer and wine and is purified into liquors.



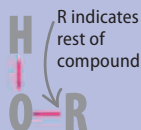
△ Propan-2-ol

This compound is so named because the functional group is on the second carbon atom.



◁ Phenol

Phenol is acidic because the hydrogen in the hydroxyl easily breaks off and reacts (see page 144).

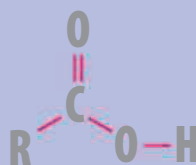


◁ Hydroxyl

The functional group with a hydrogen atom and an oxygen atom is called a hydroxyl.

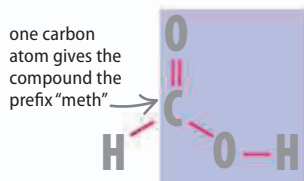
Carboxylic acids

Organic acids have a carboxyl group (COOH). The hydrogen breaks off and reacts with alkali compounds and metals. The rest of the molecule forms a carboxylate ion with a charge of –1. The salts produced when the acid reacts are called carboxylates. Most carboxylic acids are weak and have a maximum pH of around 3 or 4.



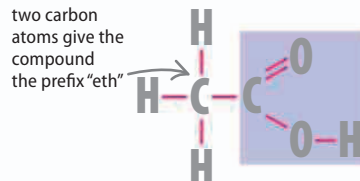
◁ Carboxyl

The carboxyl group is formed from the carbon at the end of a chain joined to one oxygen atom with a double bond and to a hydroxyl group with a single bond.



△ Methanoic acid

Also known as formic acid, this is the simplest carboxylic acid. It is used to soften animal hides into leather.



△ Ethanoic acid

Also known as acetic acid, this is the sour-tasting ingredient in vinegar. It forms naturally from ethanol due to the action of bacteria.

CFCs or chlorofluorocarbons, the chemicals that damage the ozone layer, are organic halide compounds.